MODEL CARD / VERSION 2.0 / ACADEMIC PROTOTYPE / 20 SEPTEMBER 2026

Automated Financial Customer Complaint Classification

TF-IDF + balanced Linear SVM (C=1) | Nine product categories | Human-reviewed routing only

TEST ACCURACY 88.90% | **MACRO-F1 0.6834** | **WEIGHTED-F1 0.8842**

Data and scope

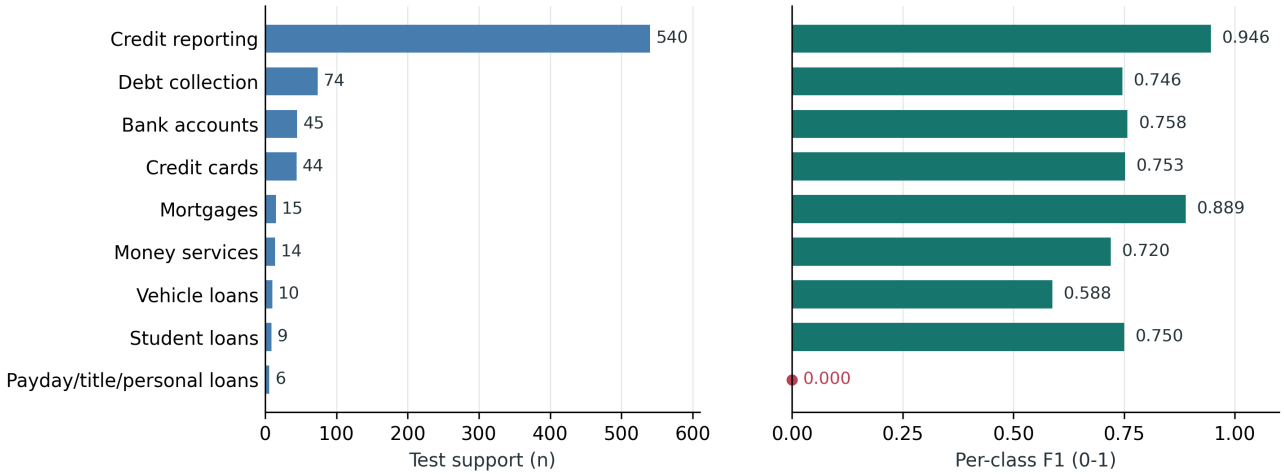
CFPB: first 10,000 API records, retrieved 13 Sep 2026; recorded dates 26 Aug-4 Sep 2024. Not a random database sample. Filtering retained **5,042**: **4,724 duplicates**, 190 conflicting-label records, 7 texts with fewer than 3 cleaned words, and 37 rare-label records removed (22 prepaid cards; 15 debt/credit management). **Empty/missing fields removed: 0**; the API request already required narratives.

Training and intended use

English-text routing suggestions, not decisions on eligibility, redress or complaint validity. Cleaned word unigrams/bigrams; TF-IDF fitted on training only (max 30,000 features). Stratified train/validation/test: 3,529/756/757, seed 42. Selected by validation macro-F1 against logistic regression, random forest and baseline. Test accuracy 95% bootstrap interval: 86.66%-91.02%.

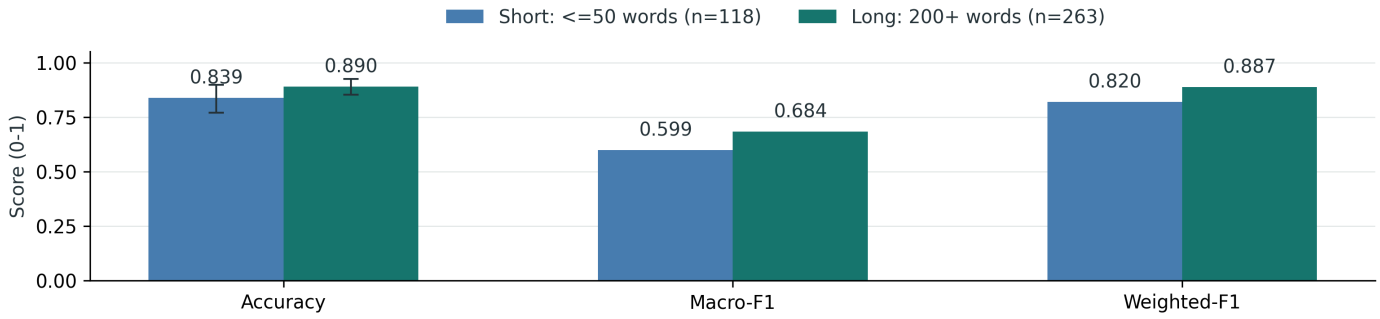
Class imbalance hides weak minority-category performance

Weighted-F1 0.8842 | Macro-F1 0.6834 | Credit reporting: 540/757 cases (71.33%)



Weighted-F1 weights classes by support; Macro-F1 gives each class equal weight. The dominant class scores well, while payday/title/personal loans has F1=0 and recall=0 (n=6).

Short complaints have lower observed accuracy



Accuracy whiskers: 95% bootstrap intervals. Length is not formality; medium texts omitted from this comparison.

Length gap: long minus short = **5.08 percentage points**; not a causal finding. Formal-detailed proxy: 85.84% (n=113); informal-short: 80.00% (n=5). Style comparison is inconclusive. Human-rated formality and demographic fairness have not been established.

Outputs, explainability and human triage

Output: highest-scoring category. Raw SVM scores/margins are **uncalibrated, not probabilities**. Explanations use feature weights and TF-IDF x weight contributions (not SHAP). **Current code:** rejects <3 cleaned words or no known vocabulary; otherwise predicts without a margin threshold. **Proposed, not implemented/validated:** general human queue for invalid input, top score <=0.10, or top-minus-runner-up gap <=0.10; otherwise category-specific human review. These illustrative cutoffs require validation-set tuning; no autonomous routing is approved.

Governance: Keep human oversight and narrative privacy controls; validate broader/temporal samples, annotate style, and monitor class/subgroup errors before deployment. Thresholds are documentation proposals, not measured safeguards.

Evidence: outputs/{classification_report,audit}.csv; test_metrics.json; data_manifest.json; data/provenance.json; src/project.py. Filtering counts recomputed from the delivered CSV. Charts use saved test results; no retraining.